

Using Deep Learning to Map Flood Damage from Aerial Imagery

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When a major hurricane or flood event strikes, emergency responders face a critical challenge: they need to know which areas are underwater before they can safely deploy search-and-rescue teams. Traditionally, assessing flood damage requires human analysts to manually review hundreds of aerial photographs from drones and satellites, which is a process that can take hours or even days. In high-stakes disaster scenarios, this delay costs lives.

Recent advances in deep learning have opened the door to a faster solution. Semantic segmentation models, the same class of computer vision models used in self-driving cars, can be trained to examine an aerial photograph and label each pixel as either flooded or non-flooded. If these models can reliably detect flood damage from overhead imagery, they could compress hours of human analysis into seconds of automated inference.

The FloodNet dataset, released by researchers at the University of Massachusetts Lowell, provides exactly the kind of labeled UAV imagery needed to train and test such a model. Collected in the aftermath of Hurricane Harvey, the dataset contains hundreds of high-resolution aerial images paired with pixel-level annotations identifying flooded buildings, flooded roads, open water, and dry land. It represents one of the most realistic and challenging benchmarks available for disaster-scene understanding.

As a data scientist working on a disaster response team, you have been tasked with building and evaluating a deep learning segmentation model on this data. Three architectures have emerged as strong candidates in recent computer vision research: U-Net, a classic encoder-decoder model designed for biomedical image segmentation; DeepLabV3+, a convolutional model that uses dilated convolutions to capture multi-scale context; and SegFormer, a transformer-based model that uses self-attention to model long-range dependencies across the image. Your job is to determine which of these approaches best handles the challenges of flood detection, including the heavy class imbalance between flooded and non-flooded pixels, and to document your findings in a reproducible GitHub repository.

The Deliverable:

In relevance to the above, you are asked to train at least one deep learning segmentation model on the FloodNet dataset and evaluate its ability to detect flooded pixels from UAV aerial imagery. Your analysis should include exploratory data analysis of the dataset, appropriate preprocessing, model training using a loss function suited to class-imbalanced data, and evaluation using segmentation metrics (IoU and F1-score). The

results you produce will illustrate the potential of automated computer vision for disaster response applications. Your final submission should be a well-documented GitHub repository containing all code, outputs, and documentation.